
MATHEMATICAL ANALYSIS 2

Improper integrals (supplementary exercises)

 **Problem 1.** Compute $\int_0^{\infty} \frac{x^2 + 1}{x^4 + 1} dx$

 **Problem 2.** Show that $\int_0^{\infty} \frac{\ln x}{x^2 + 1} dx$ is convergent and compute its value.

 **Problem 3.** Show that $\int_0^{\pi/2} \ln(\sin x) dx$ is convergent.

 **Problem 4.** $\int_0^1 \frac{\ln(1-x)}{x} dx$

 **Problem 5.** Study the convergence of $\int_0^1 \frac{\ln x}{1-x} dx$.

 **Problem 6.** Compute $\int_0^1 \frac{\ln x}{\sqrt{1-x^2}} dx$.

 **Problem 7.** Compute $\int_0^1 \frac{\arcsin x}{x} dx$.

 **Problem 8.** Compute $\int_0^{\infty} \frac{\ln(1+x^2)}{1+x^2} dx$.

 **Problem 9.** Compute $\int_0^{\infty} \arcsin e^{-x} dx$.